

Hands-on Activity 4.2 Creating Engaging Transitions and Animations	
Course Code: CPE 025B	Program: BSCPE
Course Title: Emerging Technologies in CPE 2	Date Performed: 02/16/2026
Section: CPE32S4	Date Submitted: 02/16/2026
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1. Discussion	
This module tackles the different interactivity options you can use in your website through animations and transitions. This introduces movement and motion in the website when there is an interaction between the user. This is mainly split into two which is the transition and animations. For transitions this is a concept used by a smooth change of state value over time or duration set. This is commonly seen through hover effects and navigation links which shows interaction towards the movement of the user on the webpage. Next is the use of animations and keyframes which is a multi step sequence which typically runs automatically using the @keyframe rule which is a script for the animation made. Mainly this topics focuses on promoting motion and interactivity on the webpage which is important for improving the overall aesthetic of your website.	
3. Procedure	
Scenario	
You are developing a promotional webpage for a new product launch. The goal is to create an interactive and visually appealing page that captures the attention of visitors. You'll use CSS transitions and animations to enhance the user experience and highlight key features of the product. We have provided a basic HTML skeleton for your promotional page in the editor under the <i>index.html</i> file. Your task is to apply the necessary CSS styles to this HTML structure to add engaging transitions and animations.	
Instructions	
Follow the instructions below to add transitions and animations to the webpage step by step:	
1. Apply a Fade-In Transition to the Header	
Add a fade-in transition effect to the <code>header</code> element. Set the opacity to transition from 0 to 1 over 2 seconds.	
2. Create a Button with a Color Change and Scale Animation	
Style a button element and add a hover effect that changes its background color and scales it up. Use the <code>transform</code> property to scale the button and the <code>transition</code> property to animate the changes. Also, add a smooth color transition.	

3. Apply a Keyframe Animation to the Product Image

- Create a keyframe animation that rotates the product image. Define keyframes for a smooth rotation from 0 to 360 degrees.
- Apply the animation to the `.product-image` element, setting the duration to 10 seconds and the animation to loop infinitely.

4. Animate the Feature List

- Add a slide-in animation to the feature list items. Create keyframes that move the items from left to right.
- Apply the animation to the `.feature-item` elements, staggering the start times for a sequential effect.

5. Review and Test

- Review your HTML and CSS to ensure all elements are styled correctly and the CSS rules are applied as expected.
- Experiment with different styles and properties to see how they affect your webpage.

4. Output

HTML CODE

HTML

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Product Launch</title>
  <link rel="stylesheet" href="output.css">
  <script>
    document.addEventListener('DOMContentLoaded', function() {
      document.querySelector('header').classList.add('visible');
    });
  </script>
</head>
<body>
  <header>
    <h1>Introducing Our New Product</h1>
  </header>
  <main>
    <section class="product-intro">
```

```

        
        <button class="buy-now">Buy Now!</button>
    </section>
    <section class="features">
        <h2>Features</h2>
        <ul>
            <li class="feature-item">Feature 1: High Quality</li>
            <li class="feature-item">Feature 2: Affordable
Price</li>
            <li class="feature-item">Feature 3: Easy to Use</li>
        </ul>
    </section>
</main>
<footer>
    <p>© 2024 Product Launch</p>
</footer>
</body>
</html>

```

CSS CODE

```

CSS
/* Base styles */
body {
    font-family: Arial, sans-serif;
    margin: 0;
    padding: 0;
}

header {
    background-color: #333;
    color: white;
    padding: 20px;
    text-align: center;
    opacity: 0;
    transition: opacity 2s ease-in-out;
}

header h1 {
    margin: 0;
}

header.visible {
    opacity: 1;
}

```

```
}

.product-intro {
  text-align: center;
  padding: 50px 20px;
}

.product-image {
  width: 300px;
  animation: rotate 10s infinite linear;
}

@keyframes rotate {
  from {
    transform: rotate(0deg);
  }
  to {
    transform: rotate(360deg);
  }
}

.buy-now {
  background-color: #4CAF50;
  color: white;
  padding: 15px 30px;
  border: none;
  cursor: pointer;
  transition: background-color 0.3s ease, transform 0.3s ease;
}

.buy-now:hover {
  background-color: #45a049;
  transform: scale(1.1);
}

.features {
  padding: 20px;
}

.features h2 {
  text-align: center;
}

.features ul {
  list-style: none;
  padding: 0;
}

.feature-item {
  background-color: #f4f4f4;
```

```
margin: 10px 0;
padding: 10px;
border: 1px solid #ddd;
transform: translateX(-100%);
animation: slideIn 0.5s forwards;
}

@keyframes slideIn {
  to {
    transform: translateX(0);
  }
}

.feature-item:nth-child(2) {
  animation-delay: 0.5s;
}

.feature-item:nth-child(3) {
  animation-delay: 1s;
}

footer {
  background-color: #333;
  color: white;
  text-align: center;
  padding: 10px 20px;
}
```

OUTPUT

Introducing Our New Product



Buy Now

Features

Feature 1: High Quality

Feature 2: Affordable Price

Feature 3: Easy to Use

© 2024 Product Launch

5. Supplementary Activity

HTML CODE

```
HTML
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width,
initial-scale=1.0">
  <title>Super Vehicles Marketplace</title>
  <link rel="stylesheet" href="supple.css">
</head>
<body>
  <header>
    <h1>Super Marketplace</h1>
    <nav>
      <ul class="navbar">
        <li><a href="#">Home</a></li>

        <li class="dropdown">
          <a href="#">Vehicles &#9662;</a>
          <ul class="dropdown-menu">
            <li><a href="#cars">Cars</a></li>
            <li><a href="#bikes">Bikes</a></li>
          </ul>
        </li>

        <li class="dropdown">
          <a href="#">Dealership &#9662;</a>
          <ul class="dropdown-menu">
            <li><a href="#">Manila</a></li>
            <li><a href="#">Pasay</a></li>
            <li><a href="#">Taguig</a></li>
          </ul>
        </li>

        <li><a href="#">About Us</a></li>
      </ul>
    </nav>
  </header>

  <main>
    <section id="cars" class="vehicle-section">
      <h2>Supercars</h2>
      <div class="vehicle-cards">
        <div class="card">
          
          <div class="card-content">
            <h3>Lamborghini Aventador</h3>
            <p>Top Speed: 350 km/h</p>
```

```
        <p>0-100 km/h: 2.9 sec</p>
        <p>Price: $500,000</p>
        <button>View Details</button>
    </div>
</div>

<div class="card">
    
    <div class="card-content">
        <h3>Bugatti Chiron</h3>
        <p>Top Speed: 420 km/h</p>
        <p>0-100 km/h: 2.4 sec</p>
        <p>Price: $3,000,000</p>
        <button>View Details</button>
    </div>
</div>

<div class="card">
    
    <div class="card-content">
        <h3>Ferrari F8 Tributo</h3>
        <p>Top Speed: 340 km/h</p>
        <p>0-100 km/h: 2.9 sec</p>
        <p>Price: $330,000</p>
        <button>View Details</button>
    </div>
</div>

<div class="card">
    
    <div class="card-content">
        <h3>McLaren 720S</h3>
        <p>Top Speed: 341 km/h</p>
        <p>0-100 km/h: 2.8 sec</p>
        <p>Price: $300,000</p>
        <button>View Details</button>
    </div>
</div>

<div class="card">
    
    <div class="card-content">
        <h3>Porsche 911 Turbo S</h3>
        <p>Top Speed: 330 km/h</p>
        <p>0-100 km/h: 2.7 sec</p>
        <p>Price: $220,000</p>
        <button>View Details</button>
    </div>
</div>
</div>
```

```
</section>
```

```
<section id="bikes" class="vehicle-section">
  <h2>Superbikes</h2>
  <div class="vehicle-cards">
    <div class="card">
      
      <div class="card-content">
        <h3>BMW S1000RR</h3>
        <p>Top Speed: 305 km/h</p>
        <p>0-100 km/h: 3.1 sec</p>
        <p>Price: $28,000</p>
        <button>View Details</button>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Kawasaki Ninja H2</h3>
        <p>Top Speed: 330 km/h</p>
        <p>0-100 km/h: 2.7 sec</p>
        <p>Price: $30,000</p>
        <button>View Details</button>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Ducati Panigale V4</h3>
        <p>Top Speed: 299 km/h</p>
        <p>0-100 km/h: 3.0 sec</p>
        <p>Price: $28,000</p>
        <button>View Details</button>
      </div>
    </div>

    <div class="card">
      
      <div class="card-content">
        <h3>Honda CBR1000RR-R</h3>
        <p>Top Speed: 299 km/h</p>
        <p>0-100 km/h: 3.0 sec</p>
        <p>Price: $27,000</p>
        <button>View Details</button>
      </div>
    </div>
  </div>
</div>
```



```
        </section>
    </main>

    <footer>
        <p>© 2026 Super Vehicles Marketplace</p>
    </footer>
</body>
</html>
```

CSS CODE

```
CSS

* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
    font-family: 'Arial', sans-serif;
}

body {
    background-color: #121212;
    color: #fff;
}

header {
    padding: 20px;
    background-color: #1f1f1f;
    display: flex;
    justify-content: space-between;
    align-items: center;
}

.navbar {
    display: flex;
    list-style: none;
    gap: 20px;
}

.navbar > li {
    position: relative;
}
```

```
.navbar a {
  text-decoration: none;
  color: #fff;
  padding: 10px 15px;
  display: block;
  transition: color 0.3s;
}

.navbar a:hover {
  color: #00d8ff;
}

.dropdown-menu {
  display: none;
  position: absolute;
  top: 100%;
  left: 0;
  background-color: #1f1f1f;
  min-width: 150px;
  border-radius: 5px;
  overflow: hidden;
  z-index: 100;
}

.dropdown-menu li {
  width: 100%;
}

.dropdown-menu a {
  padding: 10px;
  color: #fff;
}

.dropdown-menu a:hover {
  background-color: #333;
  color: #00d8ff;
}

.dropdown:hover .dropdown-menu {
  display: block;
}

.car-cards {
  display: flex;
  flex-wrap: wrap;
  gap: 30px;
  justify-content: center;
}
```

```
        padding: 40px;
    }

    .card {
        background-color: #1f1f1f;
        width: 300px;
        border-radius: 10px;
        overflow: hidden;
        box-shadow: 0 8px 20px rgba(0,0,0,0.5);
        transition: transform 0.3s, box-shadow 0.3s;
    }

    .card:hover {
        transform: translateY(-10px);
        box-shadow: 0 15px 30px rgba(0,0,0,0.7);
    }

    .card img {
        width: 100%;
        height: 200px;
        object-fit: cover;
        transition: transform 0.3s;
    }

    .card:hover img {
        transform: scale(1.05);
    }

    .card-content {
        padding: 20px;
    }

    .card-content h3 {
        margin-bottom: 10px;
        color: #00d8ff;
    }

    .card-content p {
        margin-bottom: 5px;
        font-size: 0.9rem;
        color: #ccc;
    }

    .card-content button {
        margin-top: 10px;
        padding: 10px 15px;
        border: none;
        background-color: #00d8ff;
        color: #121212;
        font-weight: bold;
    }
```

```
        border-radius: 5px;
        cursor: pointer;
        transition: background-color 0.3s;
    }

    .card-content button:hover {
        background-color: #0099cc;
    }

    .vehicle-section {
        padding: 40px 20px;
    }

    .vehicle-section h2 {
        text-align: center;
        margin-bottom: 30px;
        font-size: 2rem;
        color: #00d8ff;
    }

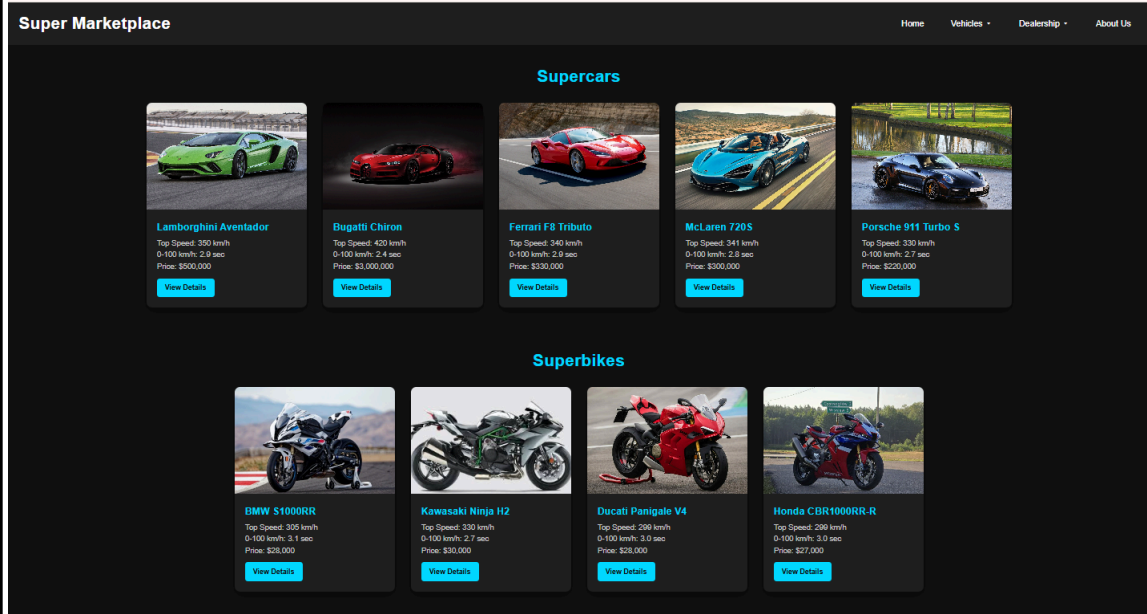
    .vehicle-cards {
        display: flex;
        flex-wrap: wrap;
        gap: 30px;
        justify-content: center;
    }

    footer {
        text-align: center;
        padding: 20px;
        background-color: #1f1f1f;
        margin-top: 40px;
        font-size: 0.9rem;
        color: #888;
    }

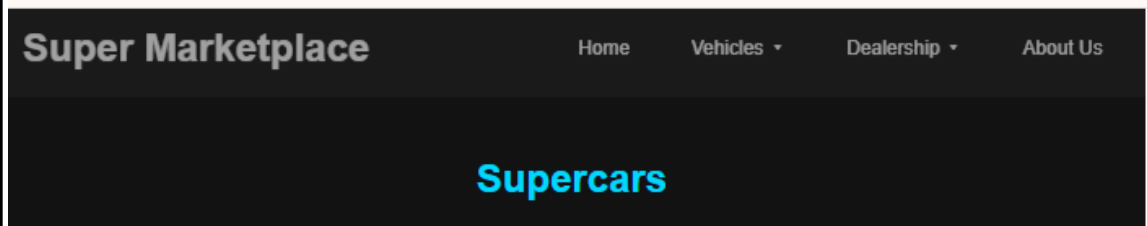
    @keyframes fadeIn {
        0% {
            opacity: 0;
            transform: translateY(-20px);
        }
        100% {
            opacity: 1;
            transform: translateY(0);
        }
    }
```

```
header {  
  animation: fadeIn 1s ease forwards;  
}
```

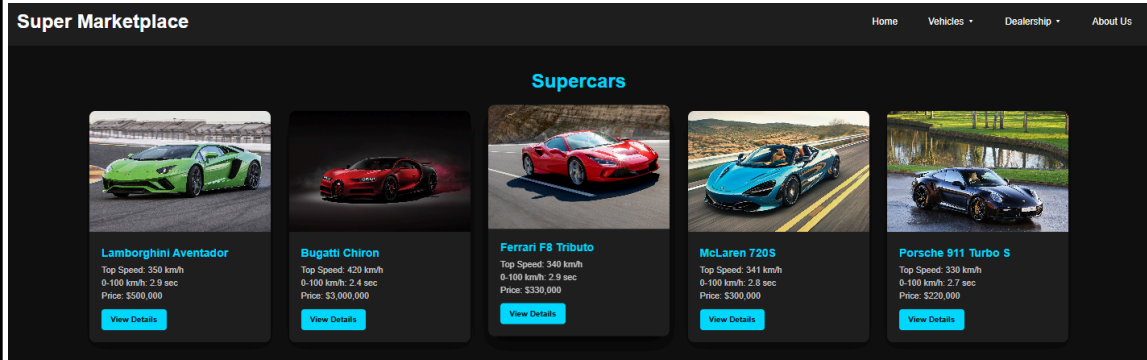
OUTPUT



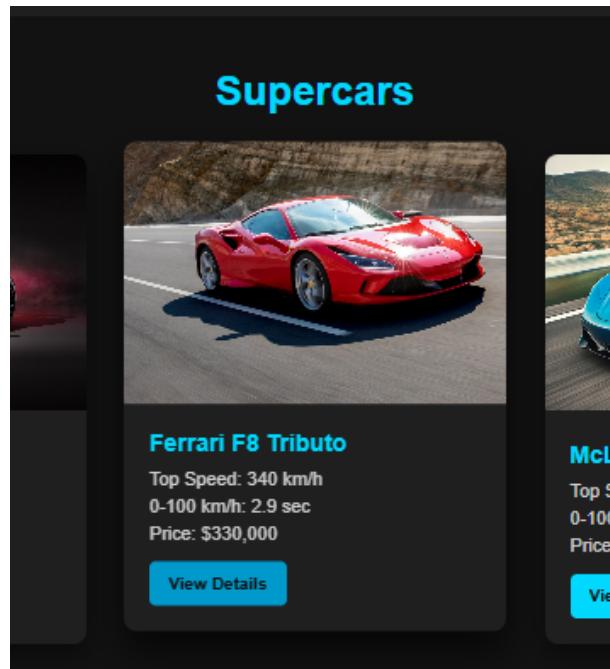
Animation on the Navigation Bar



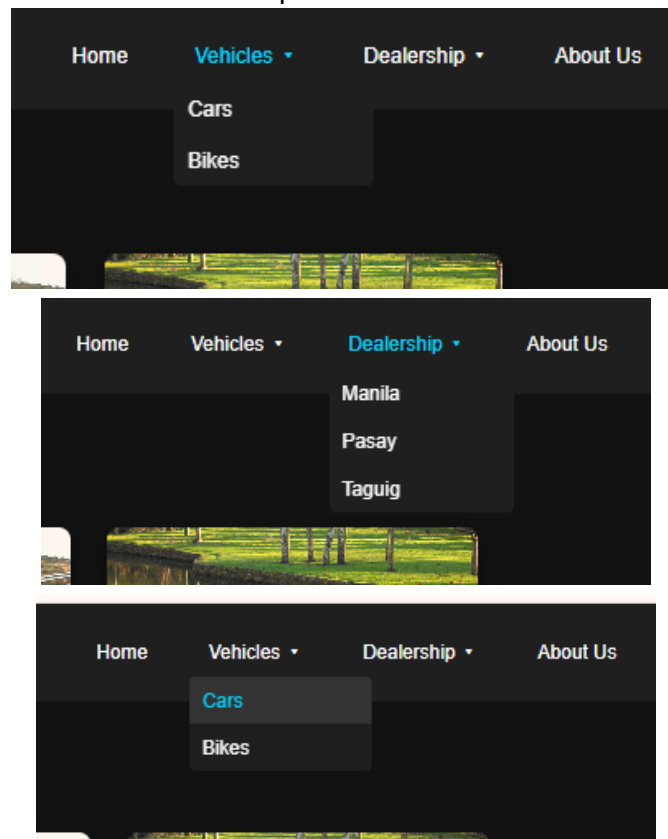
Hover Effect on Product Card



Hover Effect on Button



Drop Down Menu



CONCLUSION

After performing this activity I was able to learn about the usage of transition and animations. I learned that transitions are mainly used in hover effects and animations uses keyframes to show the motion or change in state of the element. I was able to utilize this learnings in a sample marketplace website for vehicles. In this website I focused on using transition effects mainly on the cards and the navigation bar of the website. I used the change of color and highlight and a slight movement of the movement when the mouse hovers. Then I was able to use animations only on the start of the loading of the page in the navigation bar. Overall, this activity helped me learn in implementing movement and interactivity in websites.